

Problem Statement

Travel time for a road journey depends on a mix of numerical and categorical factors — distance, average speed, number of traffic signals, traffic congestion level, weather conditions, road type, vehicle type, day of the week, and time of day. Estimating travel time accurately is essential for route planning, traffic management, and logistics scheduling.

Since several of these features are related to each other — for example, traffic level, number of signals, and average speed often move together — a plain linear regression model can suffer from multicollinearity, leading to unstable coefficients and poor generalization to new data. This project aims to build a regression model that predicts travel time (in minutes) using the available trip features, and applies Ridge and Lasso regularization to handle correlated features, improve model stability, and identify which features are most influential in determining travel time.